

Hadoop

Big data

It comes in 3 ways

Volume - quantity, variety-unstructured and velocity-speed at data come (also veracity-errors in data inconsistency)

Hadoop and spark are used to handle and deal with these big data

The frameworks which deal with big data

Mongodb

Cassandra

Hadoop

Spark

Spark is the topnotch framework written in scala

Apache spark is a general-purpose cluster in memory computing system

It is used for fast data analytics

It abstracts APIs in java, scala and python and provides an optimized engine that supports general graph

Apache Hadoop is a framework that allows for the distributed processing of large data sets

Across clusters of commodity computers using a simple programming of large model.

Framework written in java

Designed to solve problem that involve analyzing large data programming model based on google map reduce

Companies using hadoop:

Yahoo, google, facebook, amazon, aol, ibm etc

Hadoop Core Components

HDFS-hadoop distributed file system(storage)

MapReduce(processing)

Main components of HDFS

NAME NODE

DATA NODE

SECONDARY NAME NODE

Name node checks the Metadata(name.replicas,...) which will have the address and data

Data nodes contain the data

Secondary name node will be the backup of metadata

Name node

Manage file system namespace and regulates access to file by clients

metadata

The entire metadata of files maintained in memory

Metadata types

List of files

List of block of each file

List of data nodes for each block

File attributes, eg: creation time, replication factor

A transaction log

Records creation and deletion of files

Data node

One per node in the cluster, manage storage attached to the node that they can run on

A block server

Stores data in the local file system

Stores meta data of a block

Servers data and meta data to the clients

Block reports

Sends a report of all existing blocks of namenode

Secondary name node:

A misnomer. It is not a back-up in 0.21

Periodically walks up and takes checkpoints

Memory requirement are the same as name node (big)

Typically should be on a separate machine in large cluster(>10)

Directory structure is same as namenode except that keep a previous checkpoint version in addition to current

Helps in recovering a failed namenode

Usage of HDFS

SCALING WIDE

SCALING DEEP

COST EFFECTIVE

FLEXIBLE

FAULT TOLERANT

DATA BLOCK REBALANCING

ARCHITECTURE OF HADOOP

MASTER-WORKER ARCHITECTURE

There will be a hdfs client who will be interacting with client and namenode

Why are we splitting data into blocks

So the data will be distributed into block at the same time to reduce the time and cost

Pattern of block selection

Size will be 128 or multiple of 64

No of block will total size divided by block size

Communication of Datanode & NameNode

-Data node sends heartbeats every 3 seconds via a tcp handshake

-Every 10th heart beat is a block report

-namenode builds metadata from report

Communication happens through the port defined for name node

Meta data in memory

- the entire metadata is in the main memory.

- no demand paging of fs meta-data

Types of metadata

List of files

List of block of each file

List of data nodes for each block

File attributes, eg: access time, replication factor

A transaction log

- records file creations ,file deletion ,etc

Assignment week1

Lets say you have 80 tb data to store and run mapreduce on this amount of data.

Configuration of data nodes

- 8gb ram

- 100mb/s read-write speed

Total number of nodes=20

Lets assume the replication factor is 4 and block size is 64mb.

By simple calculation you will need:

=total amount of data *replication factor/total no of nodes

= $80 \times 4 / 20$

=16(disk size per datanode)

Now lets say you need to run mapreduce program on this 80 tb of data.

Reading 80tb data at aspeed of 100 MB/s using only 1 node will take:

=total data/read-write speed

= $80 \times 1024 \times 1024 / 100$

=838860.8 seconds

=13981.01 hours

With 20 data node you will be able to finish this job on

= $13981.8 / 20$

=699.05 hours

ASSIGNMENT

Take 2 csv files merge it with some usecases